Fiscal Year:	FY 2023	**Task Last Updated:**	FY 07/27/2023
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Project Title:	Measuring Pre-,In-, and Post-Flight Refractive Errors (Hyperopic Shift) in Space Adventures Flight Tourists and Crew		
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Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Comments:			
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Contact Monitor:		**Contact Phone:**	
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	EXPAND Project This grant application proposes a study of the refractive error shift first identified by Mader et al. ten years ago. Several possibilities exist for these findings, including changes in ocular axial length, changes in corneal refractive power, and changes in lens position. A reduction in axial length seems most plausible since it is consistent with the finding of a hyperopic shift and choroidal swelling. Axial length may change very soon after entering orbit, but individuals may have varying abilities to accommodate axial length changes using the ciliary body and lens. The delayed onset of symptoms may represent the point at which an individual astronaut’s accommodation has reached its limit. Alternatively, the changes in axial length may develop slowly or nonlinearly over time, perhaps representing remodeling within eye structures. Regardless, a more thorough characterization of the observed hyperopic shift would serve as a base for better understanding of the Spaceflight-Associated Neuro-ocular Syndrome (SANS) and the potential implementation of		

Task Description:	countermeasures. High-frequency ultrasound is currently used to measure axial length aboard the International Space Station (ISS). This technology is theoretically capable of detecting changes of approximately 100 microns with high accuracy (average axial-length change due to SANS-related hyperopic shift is expected to be 300 microns). For ultrasound measurements to be reproducible, the probe must be placed consistently at each measurement session. However, the literature reports previous studies wherein the observed reproducibility of obtaining ultrasound measurements was worse than 190 microns, which is likely attributed to the difficulty in obtaining consistent positioning of the probe. Building upon our previous Translational Research Institute for Space Health (TRISH) awarded project, in which we established that QuickSee's optical system could accurately measure artificially induced hyperopic shifts with sufficient reproducibility, in this project, we seek to capture relevant biometric data pertaining to refractive status of the subjects at least immediately before and after, and possibly during, spaceflight. Refractive measurements will be captured using QuickSee, a portable wavefront aberrometer which has been extensively utilized and validated in clinical studies for its accuracy, reproducibility, and ease of use. The portability of QuickSee allows for flight surgeons to take the requisite measurements quickly and accurately, in the field, at different timepoints. An advantage of the detachable QuickSee eyecup is that it can incorporate fogging lenses (e.g., +2D and +4D) which can be used to neutralize the subject's accommodation without cycloplegia, thereby bypassing the inconvenient time-course of cycloplegia, to potentially improve measurement of hyperopic shift in-flight. Furthermore, the aberrometry data captured provides not only more accurate, but also informationally-richer measurements than traditional autorefractors (i.e., high order aberrations). To the best of our knowledge, higher-order aberrometry data has not been collected in-flight, or pre-/post-flight, and may be a valuable data set. We expect to obtain quantitative data which can provide important insights on when and how fast SANS starts to be developed as well as the inter-subject variability.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The outcomes of this project have impacts both for long-duration spaceflight and terrestrial applications. 1) Enables space exploration or pioneers scientific discovery: This project has resulted in a new, feasible method to monitor to track changes in refractive error and hyperopic shift to elucidate its time course and magnitude during ground or space measurements. Future commercial spaceflights will be able to leverage these protocols and the QuickSee technology, or future versions of the technology with improved usability and clinical functionality, to collect more data to elucidate the progression (time course and magnitude) of SANS-related hyperopic shift. This type of data is particularly pertinent for long-duration spaceflight, lunar, and Martian missions, as properly functioning vision is critical to space vehicle operation. 2) Benefits for terrestrial applications to vision exams and vision monitoring: As QuickSee has already been deployed in over 40 countries and used on millions of patients worldwide over the last 4+ years, the device is already having positive impacts on patients, including those facing health disparity challenges. Projects such as the current one add to the body of clinical and scientific evaluations of the QuickSee technology, which contribute to its widescale adoption.
Task Progress:	1) Project aims/objectives and expected outcomes: To measure refractive error shift pre-, in-, and post-flight in space tourists and crew traveling on commercial flights. Deliverable 1: Provide regulatory and technical documentation for Institutional Review Board (IRB) preparation. Development of data acquisition and training protocols. Expected outcome 1. IRB approval to utilize QuickSee in the commercial space flights for pre- and post-flight testing of refractive errors. Deliverable 2 : Prepare QuickSee in research mode, remotely train flight surgeons to take QuickSee measurements, and verify training success via customized support, translations (if needed), and training materials. Expected outcome 2 : Flight surgeons successfully trained for QuickSee operation and testing protocol prior to the mission. Deliverable 3 : Download raw wavefront data from QuickSee devices, perform data analysis, store data in TrialX repository, and present the results. Expected outcome 3 : Successful acquisition, analysis, storage of raw data, and reporting of results. 2) Key findings: (1) QuickSee obtained IRB approval and flight surgeons were successfully trained remotely (Deliverables 1, 2); (2) QuickSee measurements were successfully taken on 5 research subjects across 2 separate flights (Deliverable 3); (3) Good repeatability was observed (limits of agreement/LOAs) overall; (4) The small number of research subjects made the results susceptible to outliers, which occurred with the last measurement of research subject #4; (5) The fogging lens was effective, but slightly increased the LOAs, which could be due to misalignment; and (6) The magnitude and inconsistency of the pre-flight baseline measurement timepoint relative to the actual flight (2-3 months instead of being 1-2 weeks) could mask the expected small SANS hyperopic shifts. 3) Impact of key findings on technology requirements, objectives, and specific aims of the original proposal: The QuickSee wavefront refractor and fogging eyecup achieved the objectives and specific aims of this proposal, and were successfully utilized in the field to measure refractive error shift with good repeatability. A newer version of QuickSee, which was developed with prior TRISH funding, has improved usability (reduced size, easier alignment, improved fogging lens holder) which is expected to reduce training time to the order of minutes and reduce overall time-to-measurement (from ~1 minute to ~15 seconds), thereby dramatically improving pre-, in-, and post-flight measurements of refractive error shift. 4) Proposed research plan for the coming year: QuickSee has been incorporated into other SANS-related research studies by external researchers and so we expect to continue to collect additional data.
Bibliography Type:	Description: (Last Updated: 08/11/2025)
Articles in Peer-reviewed Journals	Hernández CS, Gil A, Zaytouny A, Casares I, Poderoso J, de Lara A, Wehse A, Dave SR, Lim D, Lage E, Alejandre-Alba N. "Ametropia detection using a novel, compact wavefront autorefractor." Ophthalmic Physiol Opt. 2024 Mar;44(2):311-20. https://doi.org/10.1111/opo.13263 ; PMID: 38084770 , Mar-2024
Awards	PlenOptika. "Boston Patent Law Association "Invented Here!" Award, November 2022." Nov-2022

Awards	PlenOptika. "NE Innovation Awards (“The Eddies”), winner in “Healthy Living” track, November 2021." Nov-2021
Awards	PlenOptika. "SPIE Prism Award winner in Biomedical Devices category, January 2022." Jan-2022